

Spatial Operations

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ENGINEERING
TEXAS A&M UNIVERSITY

Overview

Vector Operations

- Spatial subsetting
- Topological relations
- Distance calculations
- Spatial joining
- Spatial aggregation

Raster Operations

- Spatial subsetting
- Map algebra
- Local operations
- Focal operations
- Zonal operations

Learning Objectives

- Apply spatial subsetting, joins, and aggregation on vectors
- Perform local, focal, zonal, and global raster operations
- Differentiate key spatial operations on vector and raster data
- Interpret spatial relationships (intersects, within, disjoint)

Practical Applications

- **Urban Planning:**
Points in polygon analysis, buffer zones, accessibility studies
- **Environmental Analysis:**
Habitat suitability, watershed analysis, land cover change
- **Remote Sensing:**
NDVI calculation, image classification, change detection
- **Emergency Response:**
Service area analysis, resource allocation, risk assessment
- **Transportation:**
Network analysis, route optimization, catchment areas

Vector Spatial Operations

Vector Spatial Operations

- Many spatial operations have a non-spatial (attribute) equivalent
 - So, concepts such as subsetting and joining datasets (demonstrated last week) are applicable here
 - This is especially true for vector operations
- Spatial operations differ from non-spatial operations in several ways
 - However spatial joins, for example, can be done in several ways while the attribution joins can only be done in one way.

Spatial Subsetting

- Selecting features based on spatial relationships
 - Process of taking a spatial object and returning only features that relate in space to another object
- Similar to attribute subsetting but uses spatial relations
- Can specify alternative operators
- Default relation: `st_intersects()`

R Syntax

```
# Basic subsetting
x[y, ]

# With specific operator
x[y, , op = st_within]

# Using st_filter()
x |> st_filter(y)
```

Topological Relations

Binary predicates describing spatial relationships

- `st_intersects()` Features that touch, cross, or are within
- `st_touches()` Features that share a boundary
- `st_overlaps()` Features that partially overlap
- `st_within()` Features completely inside another
- `st_crosses()` Features that cross each other
- `st_disjoint()` Features with no spatial relationship

Distance Relations

`st_distance()`

- Calculates continuous distance between spatial objects
- Returns values with units (meters, feet, etc.)
- Can compute distance matrices
- Supports spherical geometries (S2)
- Distance to any part of polygon

Example Usage

```
# Single distance
st_distance(point_a, point_b)

# Distance matrix
st_distance(points, polygons)

# Within distance
st_is_within_distance(
  x, y, dist = 20
)
```

Spatial Joins

`st_join()`

- Adds columns from source object to target based on spatial overlap

Types of Joins

- **Left join** (default):
keeps all target features
- **Inner join**:
only matching features
- **Distance-based**:
joins within threshold

Syntax

```
# Basic spatial join
st_join(target, source)

# With custom predicate
st_join(x, y,
        join = st_within)

# Distance-based
st_join(x, y,
        st_is_within_distance,
        dist = 100)
```

Spatial Aggregation

- Condenses data using aggregating functions (mean, sum, etc.) based on the geometry of a grouping object

Two approaches

- **Base R**, concise syntax

```
aggregate(x, by = y, FUN = mean)
```

- **tidyverse**, more flexible

```
st_join() |> group_by() |> summarize()
```

Raster Operations

Map Algebra and Spatial Processing

Map Algebra

Operations that modify or summarize raster cell values

Local	Cell-by-cell operations	Addition, subtraction, classification, Normalized Difference Vegetation Index (NDVI)
Focal	Neighborhood operations (3×3 window)	Smoothing, edge detection, slope
Zonal	Operations by categorical zones	Statistics by land cover type
Global	Entire raster operations	Min, max, mean, distance calculations

Local Operations

- Cell-by-cell operations on one or more rasters

Common Operations

- **Arithmetic:** +, -, *, /
- **Logical:** >, <, ==
- **Mathematical:** log(), sqrt(), ^2
- **Classification:** reclassify values
- **Multi-layer:** NDVI, band math

NDVI Example

```
# Define NDVI function
ndvi_fun = function(nir, red) {
  (nir - red) / (nir + red)
}

# Apply to raster layers
ndvi = lapp(
  multi_rast[[c(4, 3)]],
  fun = ndvi_fun
)
```

Focal Operations

`focal()`

- Neighborhood-based operations using moving windows
- Applies aggregation function to neighborhoods (kernels)

Applications

- **Smoothing filters** (low-pass)
- **Edge detection** (high-pass)
- **Terrain analysis** (slope, aspect)
- **Image processing**

Example

```
# Define 3x3 window
w = matrix(1, nrow=3, ncol=3)

# Apply minimum function
focal(raster,
      w = w,
      fun = min)

# Terrain processing
terrain(dem, v='slope')
```

Zonal Operations

`zonal()`

- Aggregation by categorical zones
- Computes statistics for cells grouped by zones defined in a second raster

Key Differences from Focal

- Zones **don't** have to be neighbors
- Defined by **categorical** raster
- Returns **summary table** by zone
- Can return raster with `as.raster=TRUE`

Example

```
# Calculate mean elevation by zone
z = zonal(
  elev,
  zones,
  fun = 'mean'
)
```

Key Considerations

Coordinate Reference Systems

- Both objects must have same CRS
- Reproject if necessary before operations
- Critical for accurate results

Spatial Indexing

- Uses Sort-Tile-Recursive (STR) algorithm
- Speeds up spatial queries significantly
- Automatic in `st_join()` and predicates

Memory Management

- Sparse matrices for efficient storage
- Set `sparse=FALSE` for dense output
- Consider aggregation for large datasets

Next Week

Spatial Data Operations on Vector Data

- Simplification
- Centroids
- Buffers
- Clipping
- Geometry unions

Spatial Data Operations on Raster Data

- Extent and origin
- Aggregation and disaggregation
- Resampling

Practice Question 1

Subdivision Parcel Analysis (Vector)

- Find spatial datasets for Brazos County:
 - Polygon layer containing subdivision boundaries with fields such as *subdiv_id*, *subdiv_name*
 - Point or polygon layer containing all property parcels with fields such as *parcel_id*, *land_value*, *improvement_value*, *property_type*
- Filter the parcels dataset to include only residential ones
- Perform a spatial join to estimate the residential parcels within each subdivision
 - Calculate the average land and improvement values for each subdivision
- Identify which subdivision has:
 - The highest number of residential parcels
 - The highest average land value per parcel
 - The highest average improvement value per parcel

Practice Question 2

Elevation Analysis (Raster)

- Find the Digital Elevation Model (DEM)
- Load individual DEM tiles and visualize
- Check the number of layers (bands), spatial res., CRS, extent (bounding box)
- Merge tiles into a single DEM (using the appropriate merging technique)
- Plot the “mosaicked” DEM (and save it as a new raster)
- Calculate the following terrain derivatives:
 - Slope, Aspect, Terrain Ruggedness Index (TRI), Topographic Position Index (TPI)
- Apply a 3×3 focal filter to smooth the DEM (use the min, max, and mean functions)
- Generate histograms for elevation (and slope)
 - Calculate summary statistics (min, max, mean, standard deviation)

Practice Data (Hint)

Vector

- Parcel: <https://brazoscad.org/tax-information/gis/>

Raster

- Elevation: <https://tnris.org/stratmap/elevation-lidar.html>
 - Note: Lidar Specs https://geographic.texas.gov/uploads/TxGIO_Lidar_Spec_v14.pdf

Elevation – Lidar

Main StratMap Contracts Orthoimagery **Elevation – Lidar** Hydrography Land Parcels Address Points

Lidar - Light Detection and Ranging, is a remote sensing technique that utilizes light in the form of a rapidly pulsed laser to measure return distances from the Earth captured by a sensor at the source of the pulse. These combined pulse return measurements with additional spatial and temporal data recorded by the acquisition system (airborne or terrestrial) produce a three-dimensional (3-D), detailed representation of the shape of the Earth illuminating its surface characteristics.

TxGIO acquires lidar data through partnerships with other federal and state agencies through the [StratMap Contract](#), which operates through the [Texas Department of Information Resources \(TxDIR\)](#).

TxGIO Lidar Coverage

TxGIO and partners have successfully flown Lidar to cover the entire State of Texas. All Lidar projects can be downloaded for free on the DataHub. Specifications vary by collection. Active lidar project coverage and project details (date, nominal point spacing, vendor, etc.) can be found below on the

Lidar Specifications

[Download Lidar Specifications](#)

Download Lidar

[Launch DataHub](#)

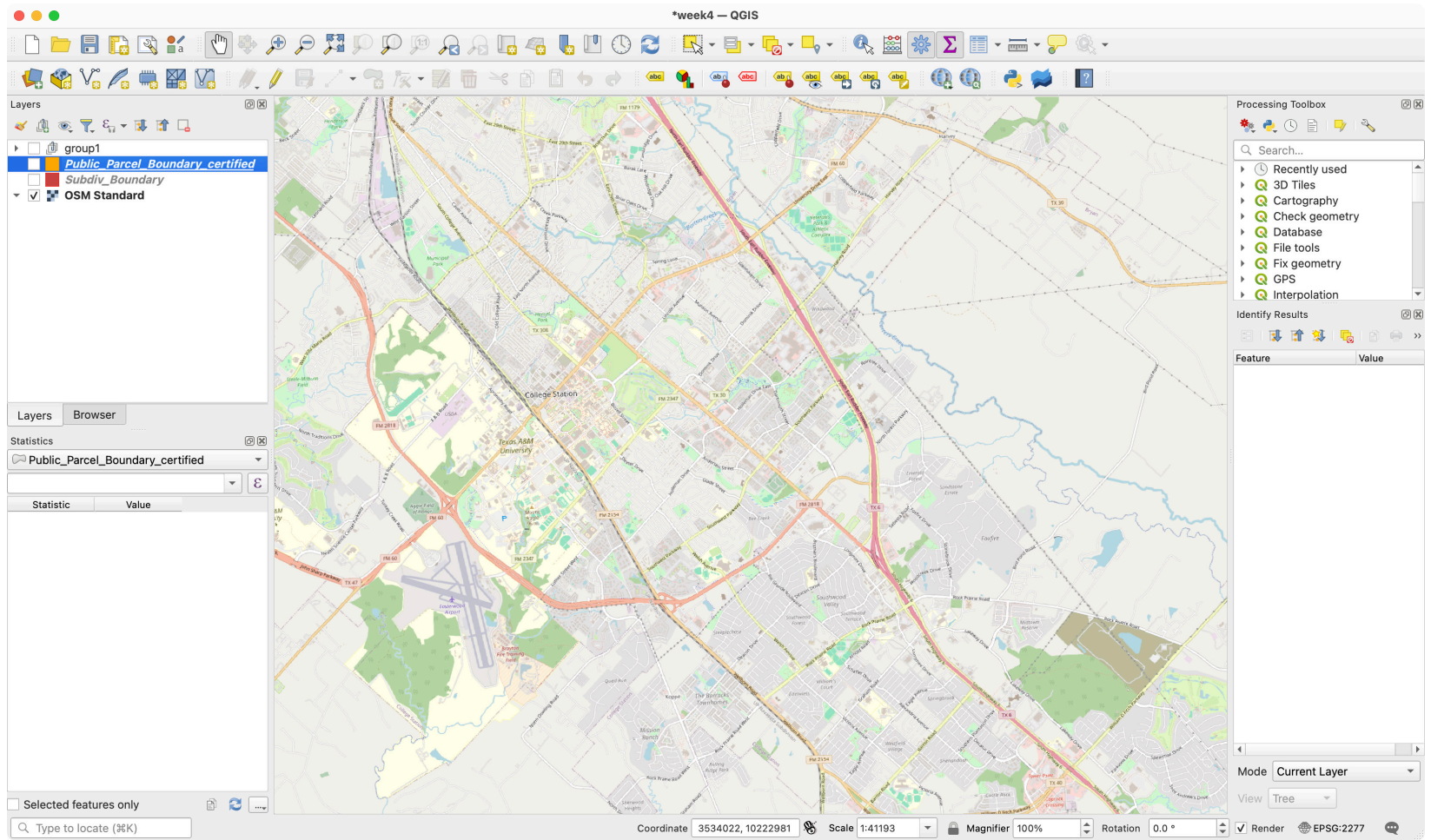
Bulk Copies

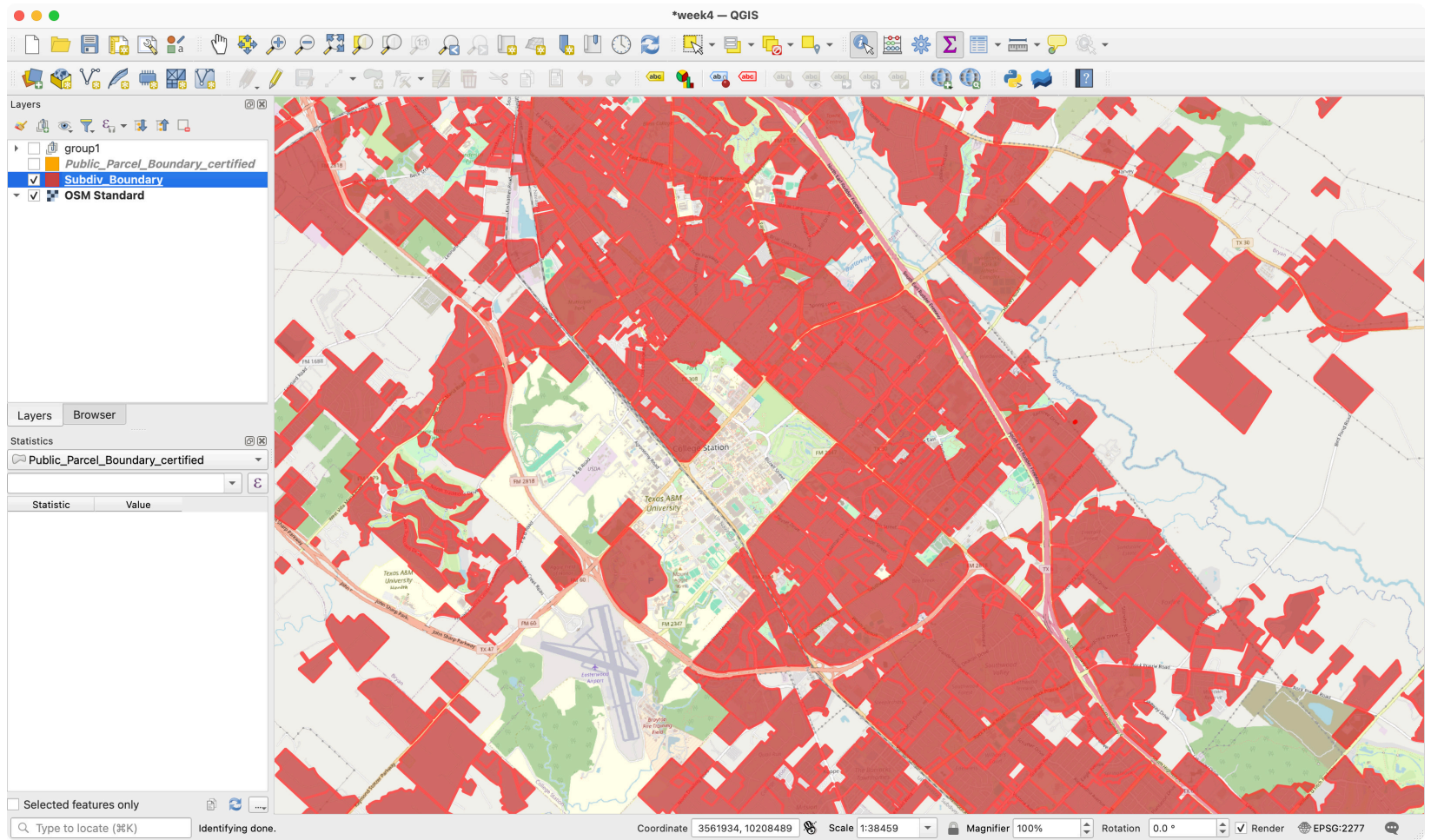
Bulk copies of all lidar datasets are available at reproduction cost through our Research and Distribution center. Copies onto hard drives can be requested through the [DataHub](#).

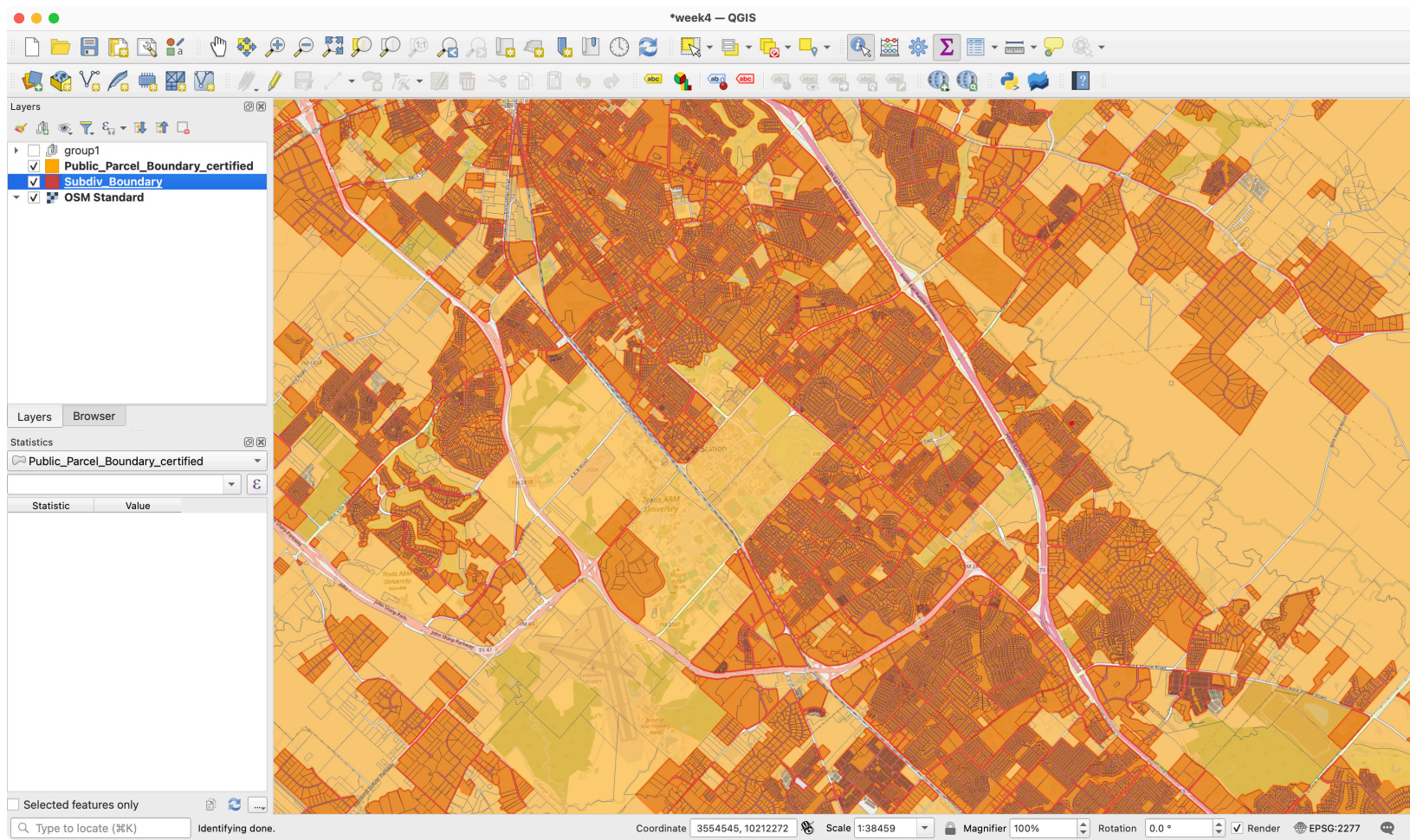
Script

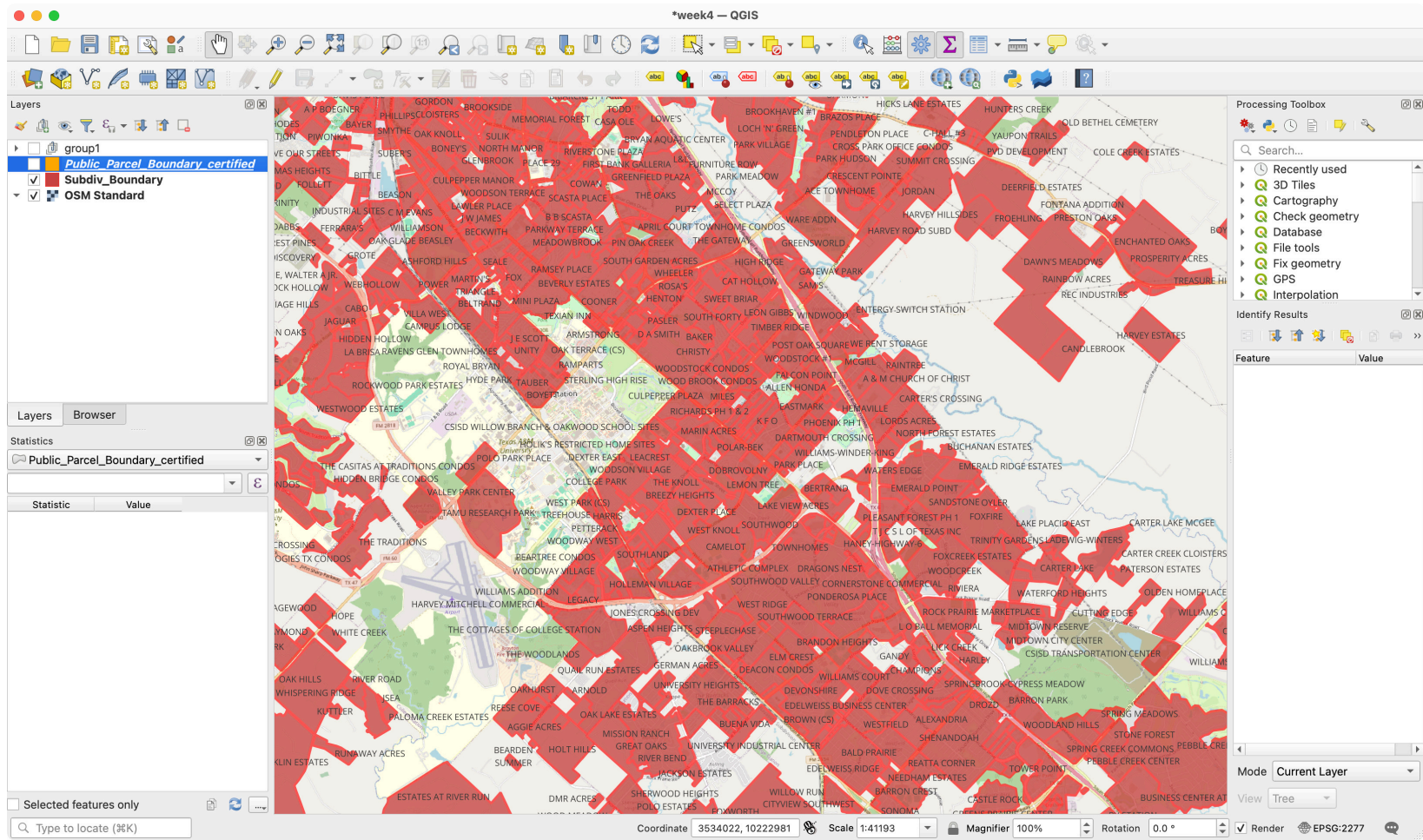
- After the in-class practice, compare your work with <https://abuabara.github.io/DAEN489S26/week4.r>

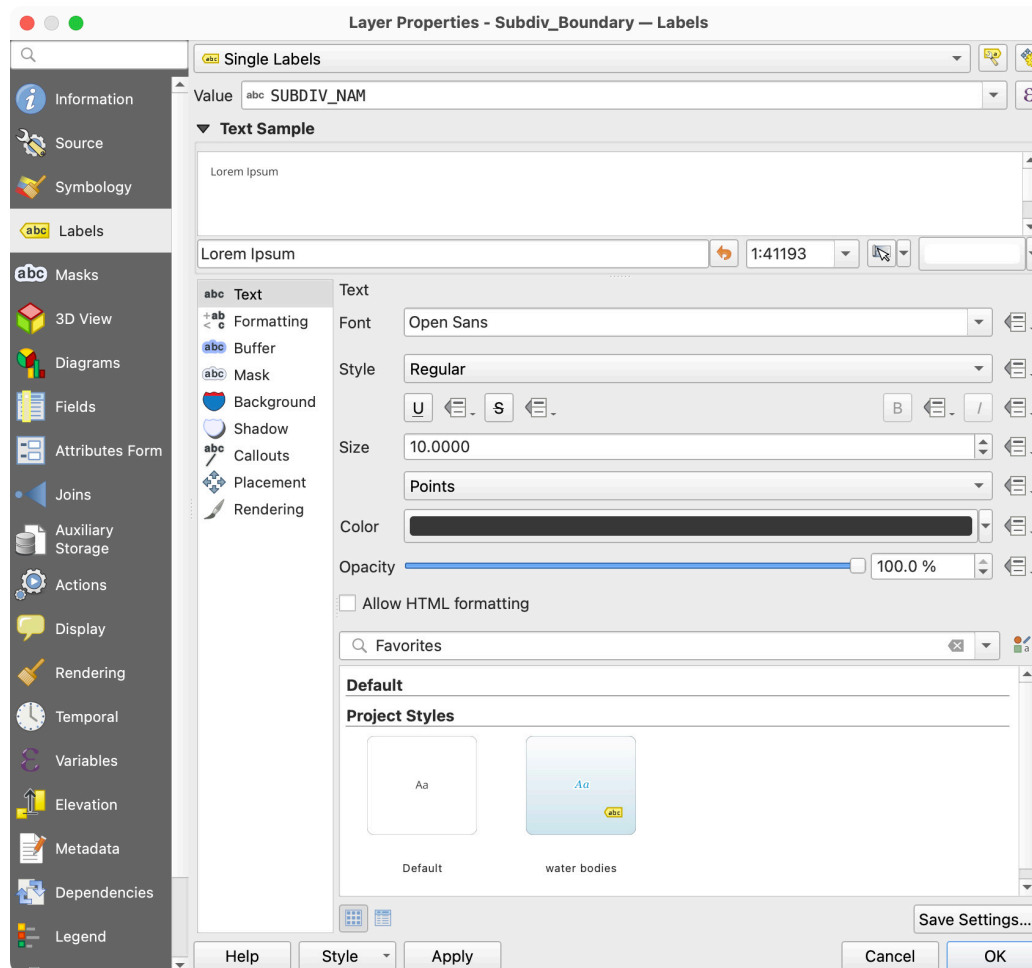
QGIS











*week4 — QGIS

Layers

- group1
- ☒ Public_Parcel_Boundary_certified
- ☒ Subdiv_Boundary
- ☒ OSM Standard

Statistics

Public_Parcel_Boundary_certified

Statistic	Value
Identifying done.	

Processing Toolbox

Search...

- Recently used
- 3D Tiles
- Cartography
- Check geometry
- Database
- File tools
- Fix geometry
- GPS
- Interpolation

Identify Results

Feature	Value
Public_Parcel_Boundary_cert	
legal_desc	RAINTREE
(Derived)	
(Actions)	
PROP_ID	38517
OBJECTID_1	23175
prop_id_1	38517
geo_id	528500-0
Sale_Date	04/02/201
sl_price	146000
sl_ratio	NULL
file_as_na	ABUABAR
addr_line1	NULL
addr_line2	8107 BUT
addr_line3	NULL
addr_city	COLLEGE
addr_state	TX
addr_zip	77845-41
situs_num	8107
situs_stre	NULL
situs_st_1	BUTLER R
situs_st_2	DR
situs_unit	NULL
legal_desc	RAINTREE
abs_subdv_	528500.0
Deed_Date	04/02/201
deed_book_	11253

Mode: Current Layer

View: Tree

Coordinate: 3570255, 10212426 Scale: 1:2423 Magnifier: 100% Rotation: 0.0 °

Render EPSG:2277

